



# REALM FORGE

## 35-Hour Hackathon Execution Plan

Mistral Worldwide Hackathon · AI-Generated Team Conquest RPG

| Total Time | Phases   | Must-Ship Features                                   | Nice-to-Have                          | Cut Entirely             |
|------------|----------|------------------------------------------------------|---------------------------------------|--------------------------|
| 35 hours   | 5 phases | World gen, map, game loop, multiplayer, voice output | Voice input (Voxtral), shareable card | Auth, mobile, animations |

## The Golden Rule

You have 35 hours. The core loop — generate world, play 3 actions, hear the DM voice — must be working by Hour 12. Everything after that is polish and features. If you are not at a working core loop by Hour 12, stop adding features and debug until it works.

| Hour 0–12                                 | Hour 12–24                          | Hour 24–32                                | Hour 32–35             |
|-------------------------------------------|-------------------------------------|-------------------------------------------|------------------------|
| Core engine: world gen + map + game logic | Multiplayer sync + ElevenLabs voice | Boss fight + polish + Voxtral voice input | Demo prep + submission |

### PHASE 1 · World Engine + Backend Setup

Hours 0 – 6 (6 hrs)

This is the most critical phase. The entire game depends on Mistral Large 3 reliably generating clean world JSON. Do not move to Phase 2 until the prompt is solid.

| Time   | Task                                                                                                                                                                        | Deliverable                             | Priority         |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|------------------|
| Hr 0–1 | Repo setup: FastAPI backend, Vite + Phaser.js frontend, folder structure, env vars, Mistral API key connected                                                               | <i>Repo runs locally, API key works</i> | <b>MUST SHIP</b> |
| Hr 1–3 | Write world_gen_system.txt prompt. Call Mistral Large 3 with 5 different world prompts. Iterate until JSON output is valid and coherent every time                          | <i>5/5 prompts return clean JSON</i>    | <b>MUST SHIP</b> |
| Hr 3–4 | Build POST /generate-world endpoint. Takes { prompt } returns full world JSON. Add basic JSON schema validation + auto-retry on malformed output                            | <i>/generate-world endpoint working</i> | <b>MUST SHIP</b> |
| Hr 4–5 | Write game_logic.py and dm_system.txt. Build POST /action endpoint. Takes { game_state, player_action } returns { narration, outcome, state_changes }                       | <i>/action endpoint working</i>         | <b>MUST SHIP</b> |
| Hr 5–6 | Write state_store.py — in-memory dict keyed by session_id. Stores full game state. Test: generate world → store it → retrieve it → apply an action → retrieve updated state | <i>State persists across calls</i>      | <b>MUST SHIP</b> |

|                      |                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>EXIT CRITERIA</b> | <i>curl /generate-world returns valid JSON world. curl /action returns valid DM narration. State updates correctly. All tests pass.</i> |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|

|               |                                                                                                                                                             |
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| <b>⚠ NOTE</b> | Spend real time on the world gen prompt here — it is the game engine. If the JSON is messy or inconsistent, everything downstream breaks. Do not rush this. |
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| <b>PHASE 2 · Map Rendering + Click Interactions</b> | <b>Hours 6 – 12 (6 hrs)</b> |
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Get a playable map on screen. Players should be able to click regions and submit actions. No multiplayer yet — single player only. Use hardcoded dummy world data to test rendering independently of the AI.

| Time   | Task                                                                                                                                      | Deliverable                              | Priority         |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|------------------|
| Hr 6–7 | Install Phaser.js. Download Kenney.nl RPG free pixel art pack. Set up WorldScene.js with a static background and basic camera             | <i>Phaser renders a canvas</i>           | <b>MUST SHIP</b> |
| Hr 7–9 | Parse world JSON → place region nodes on map at x/y coordinates. Each region is a clickable sprite with name label. Color-code by faction | <i>All regions visible and clickable</i> | <b>MUST SHIP</b> |

| Time     | Task                                                                                                                                                               | Deliverable                              | Priority         |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|------------------|
| Hr 9–10  | Build action panel UI — appears on region click. Shows region name, faction, difficulty. Has 4 action buttons: Attack, Scout, Fortify, Negotiate                   | <i>Action panel opens on click</i>       | <b>MUST SHIP</b> |
| Hr 10–11 | Add text command input box at bottom of screen. Player types natural language action. Submits to /action endpoint. DM response appears as text in a chat log panel | <i>Text commands trigger DM response</i> | <b>MUST SHIP</b> |
| Hr 11–12 | Add basic HUD: team XP bar, player role display (Warrior/Scout/Mage/Diplomat), current turn indicator. Wire XP gain from /action state_changes to HUD              | <i>HUD reflects game state</i>           | <b>MUST SHIP</b> |

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| <b>EXIT CRITERIA</b> | <i>A generated world renders on screen. Clicking a region opens action panel. Typing a command returns DM narration text. XP bar updates.</i> |
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| <b>⚠ NOTE</b> | Do not worry about making it look beautiful yet. Functional first, pretty later. Static sprites and basic shapes are completely fine at this stage. |
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| <b>PHASE 3 · ElevenLabs Voice + Full Game Loop</b> | <b>Hours 12 – 20 (8 hrs)</b> |
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This is the phase that makes the demo memorable. Adding ElevenLabs voice transforms the game from a text adventure into something that feels alive. Also completes the full win/lose loop with the boss fight.

| Time     | Task                                                                                                                                                                                               | Deliverable                                        | Priority         |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|------------------|
| Hr 12–14 | Write narration.py — takes narration text + tone from DM response → calls ElevenLabs API → returns audio URL. Assign distinct voice IDs per faction and DM narrator. Test with 5 sample narrations | <i>ElevenLabs audio plays in browser</i>           | <b>MUST SHIP</b> |
| Hr 14–15 | Wire ElevenLabs into the action pipeline. After every /action call, auto-play DM narration audio. Add a small speaker icon to mute/unmute                                                          | <i>Every action is voiced</i>                      | <b>MUST SHIP</b> |
| Hr 15–16 | Pre-generate faction leader intro voices at world gen time. When player first enters a region, faction leader speaks a short taunting line. Cache audio URLs in game state                         | <i>Faction voices play on region entry</i>         | <b>MUST SHIP</b> |
| Hr 16–18 | Build boss fight sequence. Write boss_system.txt prompt. On final region click, call Mistral Large 3 with full session history →                                                                   | <i>Boss fight triggers with adapted difficulty</i> | <b>MUST SHIP</b> |

| Time     | Task                                                                                                                                                                                                                             | Deliverable                                   | Priority         |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|------------------|
| Hr 18–19 | generates adapted boss encounter. Boss speaks opening monologue via ElevenLabs<br><br>Build victory + defeat screens. Show world name, key decisions the team made, total XP, and a world summary paragraph generated by Mistral | <i>Win/lose screens display correctly</i>     | <b>MUST SHIP</b> |
| Hr 19–20 | Full end-to-end test: world prompt → map renders → 3 actions → boss fight → victory. Fix any broken pipes                                                                                                                        | <i>Complete solo session works end-to-end</i> | <b>MUST SHIP</b> |

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| <b>EXIT CRITERIA</b> | <i>A complete solo game session works: generate world → take 3 actions with voice narration → trigger boss fight → reach victory screen.</i> |
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| <b>⚠ NOTE</b> | ElevenLabs latency can be 1–3 seconds. If it feels slow, pre-generate the most common narration phrases and cache them. Do not let latency block the demo. |
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| <b>PHASE 4 · Multiplayer Sync</b> | <b>Hours 20 – 26 (6 hrs)</b> |
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Add real-time multiplayer so 2–4 players share one game session. Keep it simple — one source of truth on the server, all clients reflect it. No conflict resolution needed; actions are queued and processed one at a time.

| Time     | Task                                                                                                                                                                                             | Deliverable                           | Priority         |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|------------------|
| Hr 20–22 | Write ws_manager.py — WebSocket connection manager. Rooms keyed by session_id. On connect, send current game state. On disconnect, remove from room. Broadcast state updates to all room members | <i>WebSocket server running</i>       | <b>MUST SHIP</b> |
| Hr 22–23 | Write wsClient.js on frontend. Connects to WebSocket on game join. Receives state updates and triggers map re-render. Shows other players' actions in chat log with player name prefix           | <i>Two tabs share live game state</i> | <b>MUST SHIP</b> |
| Hr 23–24 | Build player join flow — landing page: enter session code + choose role (Warrior/Scout/Mage/Diplomat). Session creator generates world, others join with code. Show player roster in HUD         | <i>Players can join a session</i>     | <b>MUST SHIP</b> |
| Hr 24–25 | Add action queue — only one action processes at a time. While action is processing, other                                                                                                        | <i>No concurrent action conflicts</i> | <b>MUST SHIP</b> |

| Time            | Task                                                                                                                                                                                                                                        | Deliverable                           | Priority         |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|------------------|
| <b>Hr 25–26</b> | <p>players see 'Player X is acting...' indicator. Prevents state conflicts</p> <p>Test multiplayer with 3 browser tabs. Each player takes an action. Verify all tabs update correctly. Test player disconnect/reconnect. Fix edge cases</p> | <i>3-player session works cleanly</i> | <b>MUST SHIP</b> |

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| <b>EXIT CRITERIA</b> | <i>3 browser tabs join one session, each takes actions, all tabs update in real time. No state corruption. Player disconnect is handled gracefully.</i> |
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|--------------------------------------------------|------------------------------|
| <b>PHASE 5 · Polish, Voice Input + Demo Prep</b> | <b>Hours 26 – 35 (9 hrs)</b> |
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The final stretch. This phase is split into two tracks: technical polish and demo preparation. Both matter equally. A great game demoed poorly loses. A rough game demoed brilliantly wins.

### Track A — Technical (Hours 26–32)

| Time            | Task                                                                                                                                                                                   | Deliverable                                    | Priority            |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|---------------------|
| <b>Hr 26–28</b> | Add Voxtral push-to-talk voice input. Hold button to record, release to send. Voxtral transcribes → same /action pipeline as text. Add visual recording indicator (pulsing red dot)    | <i>Voice input works via push-to-talk</i>      | <b>NICE TO HAVE</b> |
| <b>Hr 28–29</b> | UI polish pass — consistent color scheme, retro pixel font (Google Fonts: Press Start 2P), region hover effects, smoother action panel transitions, faction colors on map              | <i>Game looks intentional, not prototype-y</i> | <b>MUST SHIP</b>    |
| <b>Hr 29–30</b> | Error handling pass — what happens if Mistral API is slow? Show 'DM is thinking...' spinner. What if ElevenLabs fails? Fall back to text only. What if WebSocket drops? Auto-reconnect | <i>No crashes or silent failures in demo</i>   | <b>MUST SHIP</b>    |
| <b>Hr 30–31</b> | Add shareable victory card — on win, generate a styled summary card: world name, team name, key decisions, total session time, QR code or share link. Screenshot-able                  | <i>Shareable card generated on win</i>         | <b>NICE TO HAVE</b> |
| <b>Hr 31–32</b> | Final bug bash. Play through 3 complete sessions. Fix anything that would embarrass you in front of judges. Freeze the build — no new features after Hour 32                           | <i>3 clean sessions with no crashes</i>        | <b>MUST SHIP</b>    |

## Track B — Demo Prep (Hours 32–35)

| Time     | Task                                                                                                                                                                                                        | Deliverable                         | Priority         |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------|
| Hr 32–33 | Write and rehearse the 90-second demo script (see Section below). Time it. Practice the world prompt to use live — 'volcanic island, rival pirate clans, betrayal theme' is proven to generate great worlds | <i>Demo script memorized</i>        | <b>MUST SHIP</b> |
| Hr 33–34 | Deploy to a public URL (Render.com free tier or Google Cloud Run). Test the deployed version with two devices. Make sure WebSocket works on the deployed URL, not just localhost                            | <i>Live public demo URL working</i> | <b>MUST SHIP</b> |
| Hr 34–35 | Write the project README and submission text. Include: what it is (1 sentence), tech stack, prize categories targeted, and a link to the live demo. Rest. Eat. Do not code.                                 | <i>Submission ready to go</i>       | <b>MUST SHIP</b> |

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| <b>EXIT CRITERIA</b> | <i>Live public URL works. 90-second demo runs smoothly. Submission text written. Build frozen.</i> |
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## Master Hour-by-Hour Tracker

Print this or keep it open. Check off each row as you complete it. If you fall behind by more than 1.5 hours on any phase, skip the Nice-to-Have items in that phase and move on.

| Hours | Phase   | Key Output                                 | Status |
|-------|---------|--------------------------------------------|--------|
| 0–1   | Phase 1 | Repo + API keys connected                  |        |
| 1–3   | Phase 1 | World gen prompt iterated — 5/5 clean JSON |        |
| 3–4   | Phase 1 | /generate-world endpoint                   |        |
| 4–5   | Phase 1 | /action endpoint + DM prompt               |        |
| 5–6   | Phase 1 | State store working                        |        |
| 6–7   | Phase 2 | Phaser.js renders canvas                   |        |
| 7–9   | Phase 2 | World JSON → clickable map regions         |        |
| 9–10  | Phase 2 | Action panel on region click               |        |
| 10–11 | Phase 2 | Text command → DM text response            |        |
| 11–12 | Phase 2 | HUD with XP bar                            |        |
| 12–14 | Phase 3 | ElevenLabs audio plays                     |        |

| Hours | Phase    | Key Output                            | Status |
|-------|----------|---------------------------------------|--------|
| 14–15 | Phase 3  | Every action is voiced                |        |
| 15–16 | Phase 3  | Faction leader intro voices           |        |
| 16–18 | Phase 3  | Boss fight with adapted difficulty    |        |
| 18–19 | Phase 3  | Victory + defeat screens              |        |
| 19–20 | Phase 3  | Full solo session end-to-end test     |        |
| 20–22 | Phase 4  | WebSocket server running              |        |
| 22–23 | Phase 4  | Two tabs share live game state        |        |
| 23–24 | Phase 4  | Player join flow with session code    |        |
| 24–25 | Phase 4  | Action queue — no conflicts           |        |
| 25–26 | Phase 4  | 3-player multiplayer tested           |        |
| 26–28 | Phase 5A | Voxtral push-to-talk (nice-to-have)   |        |
| 28–29 | Phase 5A | UI polish — retro font, colors        |        |
| 29–30 | Phase 5A | Error handling — spinners, fallbacks  |        |
| 30–31 | Phase 5A | Shareable victory card (nice-to-have) |        |
| 31–32 | Phase 5A | Bug bash — 3 clean sessions           |        |
| 32–33 | Phase 5B | Demo script rehearsed                 |        |
| 33–34 | Phase 5B | Deployed to public URL                |        |
| 34–35 | Phase 5B | Submission written + REST             |        |

## The 90-Second Demo Script

Practice this out loud at least 5 times. Every word matters. Judges are tired and have seen 30 projects before yours.

| Moment     | What You Say / Do                                                                                                                        | Time   |
|------------|------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Hook       | "What if every game session was a unique world, built by AI, narrated by AI, and voiced in real time? That's Realm Forge."               | 0–10s  |
| World Gen  | Type live: 'volcanic island, rival pirate clans, betrayal theme'. Hit enter. Wait for map to appear.                                     | 10–20s |
| Wow Moment | "In about 4 seconds, Mistral Large 3 just built an entire world — factions, quests, lore, and a final boss — all unique to this prompt." | 20–30s |

| Moment      | What You Say / Do                                                                                                             | Time   |
|-------------|-------------------------------------------------------------------------------------------------------------------------------|--------|
| Action Loop | Click a region → action panel opens → type 'send scouts ahead quietly' → DM voice narrates the outcome aloud via ElevenLabs.  | 30–45s |
| Multiplayer | Switch to second tab showing teammate's screen. Take another action. Point to both screens updating simultaneously.           | 45–55s |
| Boss Tease  | Navigate to boss region. Boss's voiced opening monologue plays — it references something the team did earlier in the session. | 55–70s |
| Close       | "Different prompt, different team, different world every time. Infinite replayability, powered entirely by Mistral."          | 70–90s |

|                                                                                               |                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>NOTE</b> | Prepare a backup demo video recorded the night before. If the live demo breaks (API timeout, wifi issues), play the video. Never let technical issues kill a great project. |
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## If Things Go Wrong

| Problem                             | Fast Fix                                                                                                      |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Hour 12 and no working core loop    | Drop multiplayer entirely. Ship a polished solo experience. It's better than broken multiplayer.              |
| Mistral JSON keeps breaking         | Add a strict JSON repair function. If JSON is malformed, prompt again with 'Fix this JSON: [broken output]'.  |
| ElevenLabs latency killing the demo | Pre-generate all DM responses for your scripted demo path the night before. Cache the audio files locally.    |
| WebSocket multiplayer too buggy     | Fake multiplayer — two players use the same tab with a 'Player 1 / Player 2' toggle. No judge will know.      |
| Voxtral not working in time         | Cut it. Text + click input is fully sufficient. Voice input was always Nice-to-Have.                          |
| Running out of API credits          | Switch Minstral 8B calls to Minstral 3B for game logic (faster, cheaper). Reserve Large 3 for world gen only. |